CHENG XIANYUN

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EDUCATION

National University of Singapore (NUS) **Sep 2026 - Present**
Programme: Master of Science (Mechanical Engineering).

Central South University (CSU) & University of Dundee (UoD) **Sep 2022 - Jun 2026**
Dundee International Institute

- Major: Bachelor of Engineering in Mechanical Design, Manufacturing and Automation (First Class Honours).
- GPA: 86.33/100.
- Awards: Third-Class Scholarship and Special Scholarship, Central South University.

WORK EXPERIENCE

Member **Oct 2025 - Present**
Research Team of Yang Liu (NTU Singapore) and Tianlin Li (Beihang University)

- Developed a lightweight test-time wrapper automating routing and normalisation across 5 document formats, replacing repeated manual conversion with a unified LLM evaluation pipeline.
- Formalised metamorphic relations across 5 semantically equivalent formats and conducted 5 repeated trials across 4 LLM families, applying Jensen-Shannon divergence to quantify format-induced evidence shifts.
- Evaluated vote aggregation and format routing across 4 LLM families and 5 document formats, deriving practical model-format selection guidelines from cross-format robustness patterns.

Member **Apr 2026 - Present**
Research Team of Yang Liu (NTU Singapore)

- Assembled open-source gameplay agents into a pluggable testing framework capturing 3 data modalities, screenshots, actions and gameplay traces, with 2 interchangeable modules for execution and assessment.
- Curated reviews spanning 9 game genres and 2 reviewer types, extracting review aspects, attitudes and evidence against shared and genre-specific evaluation criteria.
- Configured persona-conditioned reviewer agents with a consistency protocol comparing 3 dimensions, evidence grounding, review focus and ratings, against experienced human players.

Team Member — TikTok TechJam 2026 **Aug 2026 – Sep 2026**
IntentCompass: Multi-turn Conversational Product Search Agent

- Analyzed multi-turn intent updates, retrieval behavior and edge-case failures for an agent-based product search system over a 50,000-item catalogue, covering buying/browsing routing, structured constraints and dynamic clarification.
- Independently validated system reproducibility on Windows, conducted failure analysis and evaluation checks, and organised evaluation evidence and the final demo workflow.

Member **Feb 2025 - Apr 2025**
UV Photodetector Design and Process Optimisation

- Specified a 3-layer Ni/n-ZnO/Ni ultraviolet photodetector with a 3-stage workflow spanning fabrication, electrical characterisation and optical validation, linking process variables to device performance.
- Fabricated ZnO films and nickel electrode patterns on sapphire through 3 microfabrication processes, RF magnetron sputtering, MA6 UV lithography and ultrasonic lift-off.
- Analysed defects associated with a 440 nm cutoff-wavelength shift and proposed sputtering and annealing adjustments to reduce defect-related spectral drift.

Member **Sep 2024 - Jan 2025**

Intelligent Firefighting Vehicle Design

- Modelled 3 mechanical subsystems, chassis, tracked drive and fire-suppression module, in SolidWorks and verified stress and deformation in ANSYS under representative loads.
- Engineered a PCB integrating 6 functional modules for motor control, power conversion, flame sensing, Bluetooth communication, relays and actuator interfaces across manual and automatic modes.
- Implemented 4 Mbed control functions for PWM speed regulation, directional movement, sensor interrupts and fan/pump actuation, then validated 3 prototype functions covering navigation, flame detection and suppression.

Member **Feb 2024 - Jun 2024**

College Student Innovation and Entrepreneurship Competition - Lightweight Virtual Try-On System

- Combined 3 vision components, MoveNet, MobileNetV2 and FPN, for real-time human keypoint extraction and pose-aware garment alignment without dedicated GPU hardware.
- Streamlined CP-VTON+ through 2 architectural modifications, depth-wise separable convolution and garment-mask input, reducing computational complexity while preserving core virtual try-on functionality.
- Applied 2 deployment optimisations, white-garment segmentation refinement and OpenVINO CPU acceleration, enabling GPU-free inference across 2 platforms, Web and PC.

INTERNSHIP EXPERIENCE

Intern **Jan 2026 - Mar 2026**

Artificial Intelligence Institute, Platform and Engineering Department, CAICT

- Researched 2 regional data ecosystems, exploring high-quality dataset development in Ningbo and data annotation practices in Shenyang, and synthesised findings for internal reporting.
- Prepared agendas and reviewed materials for 2 technical events, including WG2/WG5 joint meetings and the Fangsheng Intelligent Testing Seminar, facilitating meeting preparation and technical communication.

Intern **Oct 2025 - Dec 2025**

Changsha Jingjia Microelectronics Co., Ltd

- Analysed production-line data through 2 analysis stages, feature identification and process review, identifying process-improvement opportunities and informing engineering decisions.
- Explored across 3 semiconductor engineering areas, integrated testing and data analysis, process training and technical reporting, developing practical understanding of test coverage and post-process reliability.

STUDENT ACTIVITIES

Volunteer, 11th ICAMMT and ICMMP **Apr 2025 - May 2026**

- Managed presenter logistics and on-site coordination for 11th ICAMMT and ICMMP, assisting speakers throughout conference activities.
- Served as an interpreter and opening-ceremony emcee, facilitating communication between organisers, presenters and attendees.

Publicity Department Officer, Student Union, Dundee International Institute **Dec 2022 - Jan 2024**

- Wrote and edited 20+ WeChat articles, producing publicity content for student activities and institute communications.
- Organised practical-skills activities, including a Photoshop workshop, covering event preparation and participant coordination.

SKILLS & CERTIFICATIONS

- English: IELTS 6.5.
- Technical: Python, C++, MATLAB, SolidWorks, AutoCAD, ANSYS, Arduino, and MS Office.